ODE-FFN: Progress Report 2

Independent Study — Deep Learning

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1 Project and scope

This project asks whether an ODE-based FFN improves feature processing inside JiT, which predicts clean images from noisy images and time. The starting design composes two Taylor-flow maps around SiLU. Since Report 1, we tested its clocks, matrix parameterization, nonlinear response, readout and learned directions. Several changes help specific synthetic tasks, but no general Taylor-coupling advantage has emerged. The accompanying `attempts.md` indexes the complete recorded history, including failed settings and image work.

Most synthetic studies use a **vector residual MLP, not JiT**: width 64, three blocks, time FiLM and LayerNorm, without attention or image patches. The main comparisons below use 5,000 updates, batch 512 and AdamW at 3×10^{-4}. Training regresses velocity $X - e$ from $z_t = tX + (1-t)e$ and uniform t, using independent noise–data pairs. Gaussian, Ring8 and rotated Product16 mixtures permit analytic conditional velocity evaluation. Field MSE measures fitting; sliced Wasserstein distance (SW2) measures generated distributions. Later thin/thick experiments instead use a clean-data prediction head. Comparisons are made within each protocol.

2 What do P, the clock and Taylor order contribute?

For one original layer at $K = 2$, with $s(h) = 0.2 + 0.8\,\sigma(g(h)/10)$,

$$T(h) = h + sPh + \tfrac{1}{2}s^2P^2h, \qquad J_T = I + sP + \tfrac{1}{2}s^2P^2 + (Ph + sP^2h)\nabla s^\top. \tag{1}$$

The internal duration s is not the external flow-matching time t. The last term explains why controlling P alone does not control the full layer. In the earlier spectral-cap study, complete layer Jacobian norms decreased in all six comparisons, but SW2 improved in only three. Frozen replacement of K2 by an exponential also hurt Ring8 SW2, from 0.1848 to 0.3813. A more accurate linear solve need not preserve the function learned by a truncated model; this intervention does not compare models trained with different solvers.

Gate case study. Removing the input clock tests whether adaptation matters. Among 2,048 Ring8 samples, the fraction more than 0.6 from every mode center is 19.9% with fixed duration, 8.3% with the original input gate, and 3.4% with SwiGLU (target: 1.0%). The gate helps this separation, but adds parameters. Its P spectral norm is smaller than the fixed model's, 1.837 versus 2.139, while its full local Jacobian norm is larger, 3.176 versus 2.703. These are averages over the saved layer probes, not bounds on the whole network.

Other local controls. Freeing the effective matrices of a fixed-clock FFN lowers Ring8 field error from 0.0765 to 0.0289. Vector clocks improve Ring8 fitting but do not transfer reliably. Wider or reused clocks, biases, identity placement, augmented states, midpoint reevaluation and exact diagonal flows change individual outcomes without establishing a better quality–cost tradeoff than ordinary controls. These tests separate useful input adaptation from an assumed benefit of polynomial tying.

3 Two changes that helped the original ODE model

Complete gradient readout. We tested whether constraining the output field, rather than P alone, helps. Keeping the original ODEFFNs inside a vector network G, we predict

$$v(x,t) = \nabla_x \left[\tfrac{1}{2} x^\top G(x,t) \right] = \tfrac{1}{2} \left[G(x,t) + J_G(x,t)^\top x \right]. \tag{2}$$

This makes the complete field conservative, as is the analytic velocity of these independent Gaussian-noise mixtures. On Product16, ODE field MSE drops from 0.325527 to 0.037320. However, the same readout also helps SwiGLU192, which reaches 0.033826 at nearly equal dense inference work (182k versus 183k MACs). On Ring8, ODE reaches 0.014462 versus 0.016419 for that control; on Gaussian data, its SW2 worsens despite better field fitting. The result supports a useful computation, but not an ODE-specific benefit across tasks. The improvement over the direct parent also costs extra work.

Follow-up tests separated the transpose term from a radial derivative, removed gate derivatives, truncated reverse paths across blocks, and tried local adjoint FFNs. The full transposed derivative was useful; deleting paths also hurt ordinary models. Localizing the construction did not recover the whole-network gain. Recurrence and shared-depth controls likewise failed to establish a better ODE quality–cost tradeoff.

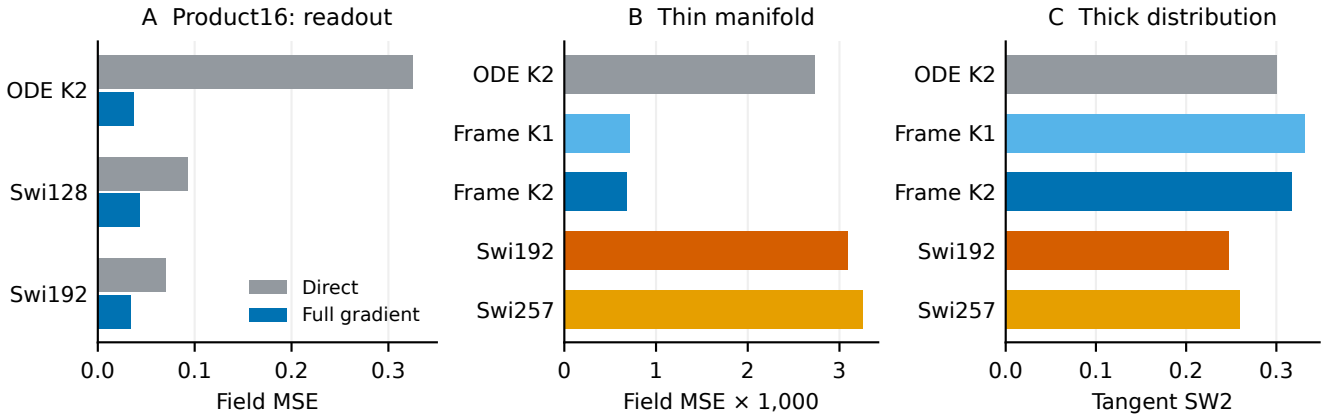


Figure 1: Lower is better. A: full-gradient readout helps Product16 field fitting for both ODE and ordinary backbones. B: conditional frames help on the thin 128D target. C: their generation advantage does not transfer to the thick target. Panels use their own metrics and scales; they are not pooled scores.

Input-dependent directions. We next replaced each original map by $Q(h)T_K(s(h)P)Q(h)^\top h$, using an orthogonal frame initialized to the identity. At a fixed input, conjugation preserves P's eigenvalues and singular values; derivatives of $Q(h)$ nevertheless change the nonlinear response. The purpose was to test directional adaptation without changing those spectra.

On eight modes in a 2D plane embedded in 128D, frame K2 reduces ambient field MSE from 0.002729 to 0.000685, versus 0.003095 for SwiGLU192. But frame K1 already reaches 0.000713, with better tangent SW2: 0.1950 versus 0.2041. SwiGLU192 nearly matches frame parameters (123.1k versus 122.9k); SwiGLU257 matches K2's dense work and also loses on this thin target. Adding variance outside the plane reverses the sampling result: frame K2's tangent SW2 is 0.3173 versus 0.2471 and 0.2593 for the two SwiGLU controls.

This is evidence for a task-dependent benefit of learned directions. It does not establish that K2 is needed. Follow-ups examined path errors, the quadratic action, an independently learned second coefficient and gradient routes. They did not establish a K2-specific advantage across both laws. The fixed output head also limits the thick model's normal-direction response; that shared bottleneck prevents attributing all thick-task error to its FFN.

4 What the other attempts rule out, and what remains

Nonlinear flow and its free-matrix control. To test whether the linear internal dynamics were too restrictive, we split $dy/dr = Py - y^{\odot 3}$ into linear steps and the exact cubic response $\phi_s(u) = u/\sqrt{1+2su^2}$. A tied split improved Product16 fitting over the original ODE, but its free control, $W_2\phi_{0.6}(W_1 h)$, did better. The free matrices train independently; they retain neither the original adaptive clock nor the generator tying.

Product16 model	Original K2	Tied cubic	Free ISRU	SwiGLU160
Field MSE	0.325527	0.092938	0.037159	0.078178

Table 1: Same 5k protocol. Free ISRU improves fitting, but it does not validate the tied ODE. The stronger SwiGLU192 direct control reaches 0.070423.

Free ISRU loses to SwiGLU160 on dependent-sign mixtures, 0.085073 versus 0.039790, and on the half mixture, 0.163557 versus 0.113254. Even on Product16, its SW2 of 0.09290 is close to the smaller SwiGLU64 control's 0.09276. The positive lead is specific to field fitting under a finite budget and a particular initialization. It should remain a control in the ODE investigation.

Locating the remaining error. We tested noisy versus analytic training labels, affine output refits, hidden features, teacher regression and local FFN response. Removing label noise did not close the ordinary-model gap. In separate frozen probes, raw-affine error accounted for about 3.9% of the direct Product16 gap; output-head refitting closed about 1%. Single-FFN teacher fits also showed an optimization gap, even without the generative objective. These observations focus attention on the learned nonlinear map and its parameterization. They are finite-budget diagnoses, not proofs of insufficient expressivity. Exact augmented-flow realizations of ordinary SwiGLU further show that an ODE interpretation alone does not distinguish our method.

Which dynamics should the architecture represent? JiT's population regression target is $D(z,t) = \mathbb{E}[X \mid Z_t = z]$, not an endpoint that must remain fixed along a sampled trajectory. For independent scalar standard Gaussian data and noise, write $V = t^2 + (1-t)^2$. Then $D = tz/V$, and along the true marginal flow its total derivative is $(1-t)z/V^2$, equal to 2 at $t = 1/2, z = 1$. Thus even a perfect denoiser changes along the trajectory. The outer ODE does not by itself justify imposing autonomous dynamics or endpoint invariance on hidden features.

Image boundary. The small CIFAR models and the larger free-ISRU study have not established an adequately trained ordinary baseline. FIDs above 200 cannot decide whether ODEFFN works. The project requires a baseline below 50 before interpreting an image architecture comparison; that is a prerequisite, not proof of convergence. Moving to author-derived JiT code addresses implementation provenance, not this missing quality evidence.

5 Hypotheses for the next two weeks

1. **Directional adaptation, rather than the quadratic term, explains the thin-target gain.** First inspect saved learning curves and interventions for frame K1/K2, the global frame and both SwiGLU controls. A targeted extension would test whether the gain survives adequate baseline training on both laws. Keep field and tangent-generation criteria separate; improvement only on thin data would support a limited geometry-specific claim.
2. **Polynomial tying offers no benefit beyond its initialization and optimization coordinates in the tested setting.** Compare a tied map with its free effective-matrix realization at the same initial function, response and budget, explicitly accounting for Adam's coordinate dependence. Use the existing static-map and optimizer-transport checks to specify this comparison. A persistent benefit must exceed both a parameter control and a compute control before another ODE design is proposed.